MH1200 Linear Algebra I.

Problem set #11.

SOLUTIONS

Problem 1:

Determine whether or not the following vectors form a basis for $\mathbb{R}^4$:

$$\mathbf{u_1} = (1, 0, -1, 2)\ , \quad \mathbf{u_2} = (2, 1, 3, 1)\ , \quad \mathbf{u_3} = (0, 5, 2, 0)\ , \quad \mathbf{u_4} = (4, 0, -4, 8)$$

Solution

We must check whether the vectors

1. span $\mathbb{R}^4$
2. and are linearly independent.

In lectures we developed a matrix algorithm to check whether they span $\mathbb{R}^4$. You make a matrix whose columns are the given vectors:

$$\begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ -1 & 3 & 2 & -4 \\ 2 & 1 & 0 & 8 \end{bmatrix}$$

Then you do Gaussian elimination. If you end up with any zero row then the vectors don't span $\mathbb{R}^4$. Otherwise they do. Gaussian elimination gives us:

$$\to \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ 0 & 5 & 2 & 0 \\ 0 & -3 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & -23 & 0 \\ 0 & 0 & 15 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 4 \\ 0 & 1 & 5 & 0 \\ 0 & 0 & -23 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The result of Gaussian elimination has a row of zeros. Thus these vectors don't span $\mathbb{R}^4$. Thus the vectors can't form a basis.

$\square$

Problem 2:

Prove that the concept of coordinate vector makes sense ("is well-defined"). To be precise show that if there are two sets of coefficients giving $\mathbf{w}$

$$\mathbf{w} = c_1\mathbf{v}_1 + \ldots + c_n\mathbf{v}_n \quad \text{and} \quad \mathbf{w} = d_1\mathbf{v}_1 + \ldots + d_n\mathbf{v}_n$$

then

$$c_1 = d_1\,,\ c_2 = d_2\,,\ \ldots\,,\ c_n = d_n.$$

So there is a unique coordinate vector attached to every vector $\mathbf{w}$.

Solution

We are told that

$$c_1\mathbf{v}_1 + \ldots + c_n\mathbf{v}_n = \mathbf{w} = d_1\mathbf{v}_1 + \ldots + d_n\mathbf{v}_n.$$

Thus, taking everything to the left hand side of the equation:

$$(c_1 - d_1)\mathbf{v}_1 + \ldots + (c_n - d_n)\mathbf{v}_n = \mathbf{0}.$$

Now because these vectors form a basis, they are linearly independent. Thus all these coefficients are zero. We deduce:

$$(c_1 - d_1) = (c_2 - d_2) = \ldots = (c_n - d_n) = 0.$$

□

Problem 3:

Determine which of the following sets are bases for $\mathbb{R}^3$. If it is a basis, find the coordinates of $(1, 2, 3)$ relative to that basis.

1. $S_1 = \{\,(1, 0, -1),\ (-1, 2, 3)\,\}$.
2. $S_2 = \{\,(1, 0, -1),\ (-1, 2, 3),\ (0, 3, 0)\,\}$.
3. $S_3 = \{\,(1, 0, -1),\ (-1, 2, 3),\ (0, 3, 3)\,\}$.
4. $S_4 = \{\,(1, 0, -1),\ (-1, 2, 3),\ (0, 3, 0),\ (1, -1, 1)\,\}$.

Solution

To begin: Let's exploit a few facts we know. Namely:

- The dimension of $\mathbb{R}^3$ is 3.
- Every basis for a subspace has the same number of elements.

Thus, any basis for $\mathbb{R}^3$ has to have 3 vectors. So the only possible bases in the list given are S_2 and S_3. We'll test these two explicitly.

The case S_2. We employ the matrix algorithms we know to test the two properties we need: (1) Do they span $\mathbb{R}^3$? (2) Are they linearly independent?

To test whether S_2 spans $\mathbb{R}^3$, the matrix algorithm tells us to start by making a matrix whose columns are the given vectors.

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ -1 & 3 & 0 \end{bmatrix}$$

Then do Gaussian elimination. If there are any zero rows then the vectors cannot span $\mathbb{R}^3$. Otherwise they do. We calculate:

$$\rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ 0 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & -3 \end{bmatrix}$$

This matrix is in row echelon form and has no zero rows. Thus the vectors span $\mathbb{R}^3$.

To test whether these vectors are linearly independent: we use the same matrix and check whether every column contains the leading entry of some row. Yes they do. (Actually this is inevitable for *any* n spanning vectors in $\mathbb{R}^n$. Why?)

In conclusion: S_2 forms a basis for $\mathbb{R}^3$.

Next we are asked to find the coordinate vector for $(1, 2, 3)$ with respect to this basis. That is, we need to solve for the unique numbers c_1, c_2, c_3 such that

$$(1, 2, 3) = c_1(1, 0, -1) + c_2(-1, 2, 3) + c_3(0, 3, 0).$$

This linear system gives the following augmented matrix which we proceed to solve:

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ -1 & 3 & 0 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 2 & 0 & 4 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & -3 & 2 \end{array}\right]$$

Back-substitution gives: $c_3 = -2/3$. Then $2c_2 + 3c_3 = 2$ which gives $c_2 = 2$. Finally we deduce $c_1 = 1 + c_2 = 3$. So the coordinate vector of $(1, 2, 3)$ is $(3, 2, -2/3)$. Let's check that

$$(1, 2, 3) = 3 * (1, 0, -1) + 2 * (-1, 2, 3) + (-2/3)(0, 3, 0).$$

This checks out.

The case S_2. In this case if we apply the algorithm this is what happens:

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ -1 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ 0 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

This row echelon form has an all-zero row, so the given list of vectors S_3 cannot span $\mathbb{R}^3$. (Actually if we had sharp eyes we might have noticed immediately that $3(1, 0, -1) + 3(-1, 2, 3) - 2(0, 3, 3) = (0, 0, 0)$ so that the list is linearly dependent.)

$\square$

Problem 4:

Let W denote the subspace of $\mathbb{R}^4$ spanned by the following vectors:

$$\mathbf{w_1} = (1, 2, -1, 1)\ , \quad \mathbf{w_2} = (4, 1, 3, -1)\ , \quad \mathbf{w_3} = (0, 0, 0, 1)$$
$$\mathbf{w_4} = (5, 3, 2, -1)\ , \quad \mathbf{w_5} = (2, 0, 2, 1)\ .$$

Identify a linearly independent subset of $\{\mathbf{w_1}\ , \mathbf{w_2}\ , \mathbf{w_3}\ , \mathbf{w_4}\ , \mathbf{w_5}\}$ which forms a basis for W. What is the dimension of W?

Solution

We developed a matrix algorithm to do this in the lectures. You make a matrix with these vectors as columns, then you do Gaussian elimination, then you take the vectors from the original list whose corresponding columns at the end of Gaussian elimination contain a leading entry:

$$\begin{bmatrix} 1 & 4 & 0 & 5 & 2 \\ 2 & 1 & 0 & 3 & 0 \\ -1 & 3 & 0 & 2 & 2 \\ 1 & -1 & 1 & -1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & -7 & 0 & -7 & -4 \\ 0 & 7 & 0 & 7 & 4 \\ 0 & -5 & 1 & -6 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 4 & 0 & 5 & 2 \\ 0 & -7 & 0 & -7 & -4 \\ 0 & 0 & 1 & \neq 0 & \neq 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

I've just written $\neq 0$ because we don't need these numbers.

In this row echelon form, the first three columns contain the leading entries of rows. Thus the first three vectors of the original list will form a basis for the span of the list. The basis is:

$$\mathbf{w_1} = (1, 2, -1, 1)\ , \quad \mathbf{w_2} = (4, 1, 3, -1)\ , \quad \mathbf{w_3} = (0, 0, 0, 1).$$

There are three vectors in this basis, so by definition, the dimension of this space is 3.

$\square$

Problem 5:

Determine the ranks of the following matrices.

$$A = \begin{pmatrix} 1 & 43 & -37 \\ 0 & -12 & 12 \\ 0 & -12 & 12 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 7 & -1 \\ 2 & 2 & 10 \\ 3 & 9 & 9 \\ 1 & 1 & 1 \end{pmatrix}.$$

Solution

The rank of a matrix can be defined in many different ways. The simplest is probably the number of non-zero rows in row echelon form. Consider first A, and put it in row echelon form.

$$A = \begin{bmatrix} 1 & 43 & -37 \\ 0 & -12 & 12 \\ 0 & -12 & 12 \end{bmatrix} \to \begin{bmatrix} 1 & 43 & -37 \\ 0 & -12 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

There are two non-zero rows, so the rank of A is 2.

Next consider B:

$$B = \begin{bmatrix} 1 & 7 & -1 \\ 2 & 2 & 10 \\ 3 & 9 & 9 \\ 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 7 & -1 \\ 0 & -12 & 12 \\ 0 & -12 & 12 \\ 0 & -6 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 7 & -1 \\ 0 & -6 & 2 \\ 0 & -12 & 12 \\ 0 & -12 & 12 \end{bmatrix} \to \begin{bmatrix} 1 & 7 & -1 \\ 0 & -6 & 3 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix}$$

There are 3 non-zero rows, so the rank is 3.

$\square$

Problem 6:

For each of the following cases, find a matrix with the stated property or explain why no such matrix exists.

(i) Column space contains the vectors $(1,0,0)$, $(0,0,1)$ and row space contains the vectors $(1,1)$, $(1,2)$.

(ii) Column space $= \mathbb{R}^4$, row space $= \mathbb{R}^3$.

Solution to (i)

First note such a matrix must be 3×2. If play around long enough we can write down

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Note that

$$(1,1) = 1(1,1) + 0(0,0) + 0(0,1) \quad \text{and} \quad (1,2) = 1(1,1) + 0(0,0) + 1(0,1)$$

so the row space condition is met. Also

$$(1,0,0) = 1(1,0,0) + 0(1,0,1) \quad \text{and} \quad (0,0,1) = 1(1,0,1) + (-1)(1,0,0).$$

Solution to (ii)

This can't happen. According to a theorem we proved in lectures, the dimension of the row space must always equal the dimension the column space. But $\mathbb{R}^4$ has dimension 4 while $\mathbb{R}^3$ has dimension 3.

$\square$

Problem 7:

Let

$$A = \begin{pmatrix} 7 & -2 & -1 \\ 4 & 1 & -3 \\ 1 & -1 & 0 \\ 8 & -3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 8 & 8 & 8 & 38 \\ -8 & 8 & 8 & 38 \\ 8 & -8 & 8 & 38 \\ 8 & 8 & -8 & -38 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 0 & -3 & -1 \\ 2 & 4 & 1 & -8 & 3 \\ -1 & -2 & -1 & 5 & -4 \end{pmatrix}$$

For each matrix, do the following:

(i) Find bases for the row space, column space and nullspace of the matrix.

(ii) From part (i) determine the rank and nullity of the matrix, and check that the Rank-Nullity theorem is satisfied.

Solution for the matrix A

To determine bases for the rank, column and null space, we start by putting the matrix into row echelon form.

$$A = \begin{bmatrix} 7 & -2 & -1 \\ 4 & 1 & -3 \\ 1 & -1 & 0 \\ 8 & -3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 7 & -2 & -1 \\ 4 & 1 & -3 \\ 8 & -3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 5 & -1 \\ 0 & 5 & -3 \\ 0 & 5 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 & 0 \\ 0 & 5 & -1 \\ 0 & 0 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

From this we can read off the various required objects.

A basis for the row space is given by:

$$\{(1, -1, 0), (0, 5, -1), (0, 0, -2)\}$$

Every column of the echelon form has a leading entry so the original columns are linearly independent. Thus a basis for the column space is given by:

$$\{(7, 4, 1, 8), (-2, 1, -1, -3), (-1, -3, 0, 2)\}$$

Thus the rank is 3.

The three columns are linearly independent. Thus the null space is trivial $\{0\}$. Thus the nullity is 0.

The dimension theorem is satisfied: Rank (3) plus nullity (0) equals the number of columns (3).

Solution for the matrix B

We transform B into echelon form:

$$B = \begin{bmatrix} 8 & 8 & 8 & 38 \\ -8 & 8 & 8 & 38 \\ 8 & -8 & 8 & 38 \\ 8 & 8 & -8 & -38 \end{bmatrix} \to \begin{bmatrix} 8 & 8 & 8 & 38 \\ 0 & 16 & 16 & 76 \\ 0 & -16 & 0 & 0 \\ 0 & 0 & -16 & -76 \end{bmatrix} \to \begin{bmatrix} 8 & 8 & 8 & 38 \\ 0 & 16 & 16 & 76 \\ 0 & 0 & 16 & 76 \\ 0 & 0 & -16 & -76 \end{bmatrix} \to \begin{bmatrix} 8 & 8 & 8 & 38 \\ 0 & 16 & 16 & 76 \\ 0 & 0 & 16 & 76 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus a basis for the row space is given by:

$$\{(8,8,8,38),(0,16,16,76),(0,0,16,76)\}.$$

The first three columns have leading entries, so the first three columns of the original matrix give a basis for the column space:

$$\{\{(8,-8,8,8),(8,8,-8,8),(8,8,8,-8)\}\}$$

The rank of the matrix B is 3.

To write down a basis for the null space we'll first write down the general solution to the equation $Bx = 0$. Using back-substitution from the row echelon form we deduce the null space is:

$$\left\{ \begin{bmatrix} 0 \\ 0 \\ -\frac{19}{4}s \\ s \end{bmatrix} : s \in \mathbb{R} \right\}$$

Thus a basis for the null space is given by the vector:

$$\begin{bmatrix} 0 \\ 0 \\ -\frac{19}{4} \\ 1 \end{bmatrix}.$$

The nullity is 1.

As expected, the dimension theorem is satisfied: Rank (3) + Nullity (1) = Number of columns (4).

<u>Solution for the matrix C</u>

Applying row reduction to C:

$$C = \begin{bmatrix} 1 & 2 & 0 & -3 & -1 \\ 2 & 4 & 1 & -8 & 3 \\ -1 & -2 & -1 & 5 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & -2 & 5 \\ 0 & 0 & -1 & 2 & -5 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & -3 & -1 \\ 0 & 0 & 1 & -2 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus the row space has a basis:

$$\{(1,2,0,-3,-1),(0,0,1,-2,5)\}.$$

The first and third columns of the row echelon form contain leading entries. So the first and third columns of the original matrix are a basis for the column space:

$$\{(1,2,-1),(0,1,-1)\}$$

The rank of the matrix is 2.

To find the null space, we must write down the general solution of the equation: $Cx = 0$. The equivalent system consists of two equations:

$$x_1 + 2x_2 - 3x_4 - x_5 = 0\ , \quad x_3 - 2x_4 + 5x_5 = 0.$$

We introduce free parameters: $x_2 = r$, $x_4 = s$, $x_5 = t$. Then we solve for x_1 and x_3 to deduce that the null space is:

$$\left\{ \begin{bmatrix} -2r + 3s + t \\ r \\ 2s - 5t \\ s \\ t \end{bmatrix} : r, s, t \in \mathbb{R} \right\}$$

Or:

$$\left\{ r \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} : r, s, t \in \mathbb{R} \right\}$$

Thus a basis for the null space is given by:

$$\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Thus the nullity is 3.

As expected the rank-nullity theorem is satisfied. Rank (2) + Nullity (3) = Number of columns (5).

$\square$

Problem 8:

Determine a basis for and state the dimension of each of the following subspaces of $\mathbb{R}^4$.

1. the subspace containing all vectors of the form $(w, 0, y, 0)$.
2. the subspace containing all vectors of the form (w, x, x, w).
3. the subspace containing all vectors of the form (w, x, y, z) with $w = 2x = 3y = 4z$.
4. the solution space of
$$\begin{cases} 2w + 3x + y + z = 0 \\ -3w + x + 4y - 7z = 0 \\ w + 2x + y = 0. \end{cases}$$
5. the subspace $\{\, (w, x, y, z) \mid y = w + x \text{ and } z = w - x \,\}$.

Solution

Preliminary comments about the solution method in general:

The way to solve problems of this type is to express the general vector as a linear combination of terms which each contain only one of the free parameters. Then write each term as free parameter multiplied onto a vector. Those vectors then span the subspace, and are therefore the candidate for a basis for the subspace. To show they really do form a basis we need to show that they are linearly independent. (And if they are not linearly independent we need to discard one or more of them, so that the vectors that remain are linearly independent and still span the subspace.)

Solution to part 1:

$$(w, 0, y, 0) = (w, 0, 0, 0) + (0, 0, y, 0) = w(1, 0, 0, 0) + y(0, 0, 1, 0)$$

Since w, y are free parameters, this shows that the subspace is $V = \operatorname{span}\{(1, 0, 0, 0)\,, (0, 0, 1, 0)\}$. This will be a basis for V if the vectors are linearly independent. To see if they are, consider

$$\mathbf{0} = x_1(1, 0, 0, 0) + x_2(0, 0, 1, 0) = (x_1, 0, x_2, 0)\,.$$

Clearly the only solution is the trivial solution $x_1 = x_2 = 0$. This shows that the vectors are linearly independent and therefore a basis for V, and so the dimension of V is 2.

Solution to part 2:

$$(w, x, x, w) = (w, 0, 0, w) + (0, x, x, 0) = w(1, 0, 0, 1) + x(0, 1, 1, 0)$$

and the only solution to

$$\mathbf{0} = x_1(1, 0, 0, 1) + x_2(0, 1, 1, 0) = (x_1, x_2, x_2, x_1)$$

is the trivial solution $x_1 = x_2 = 0$, so this subspace has the basis $\{(1, 0, 0, 1)\,, (0, 1, 1, 0)\}$, hence its dimension is 2.

Solution to part 3: The constraint $w = 2x = 3y = 4z$ can be re-expressed as $x = w/2$, $y = w/3$, $z = w/4$, with w being the only free parameter. Then we have

$$(w, x, y, z) = (w, w/2, w/3, w/4) = w(1, 1/2, 1/3, 1/4)$$

showing that the subspace has the basis $\{(1, 1/2, 1/3, 1/4)\}$ and its dimension is 1.

Solution to part 4: Solving the linear system in the usual way by Gaussian elimination and back substitution, the general solution is found to be $(w, x, y, z) = (s - 2t, -s + t, s, t)$ where s and t are free parameters. We re-express this as

$$(w, x, y, z) = (s, -s, s, 0) + (-2t, t, 0, t) = s(1, -1, 1, 0) + t(-2, 1, 0, 1).$$

The only solution to

$$\mathbf{0} = x_1(1, -1, 1, 0) + x_2(-2, 1, 0, 1) = (x_1 - 2x_2\,, -x_1 + x_2\,, x_1\,, x_2)$$

is the trivial solution $x_1 = x_2 = 0$. This shows that $\{(1, -1, 1, 0)\,, (-2, 1, 0, 1)\}$ is a basis for the solution space, which therefore has dimension 2.

Solution to part 5: With the constraints $y = w + x$ and $z = w - x$ we have

$$(w, x, y, z) = (w, x, w + x, w - x) = (w, 0, w, w) + (0, x, x, -x) = w(1, 0, 1, 1) + x(0, 1, 1, -1).$$

The only solution to

$$\mathbf{0} = x_1(1, 0, 1, 1) + x_2(0, 1, 1, -1) = (x_1, x_2, *, *)$$

is $x_1 = x_2 = 0$. This shows that $\{(1, 0, 1, 1)\,, (0, 1, 1, -1)\}$ is a basis for the subspace, which therefore has dimension 2.

$\square$

Problem 9.

Assume that $\mathbf{A}$ is a 100×101 matrix whose rank is 100. Do there exist any 100×1 matrices $\mathbf{C}$ such that the equation

$$\mathbf{AX} = \mathbf{C}$$

has a unique solution $\mathbf{X}$? Justify your answer.

Solution version 1

Because $\mathbf{A}$ has more columns than rows, in the RREF there will always be a column without a leading entry. So the only two possibilities are that the system is inconsistent or that the set of solutions has a free parameter, and hence infinite solutions.

Solution version 2

By the rank-nullity theorem we deduce that:

$$\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = 101.$$

Thus:

$$\operatorname{nullity}(\mathbf{A}) = 101 - 100 = 1.$$

Thus there exists a non-zero matrix $\mathbf{B}$ with the property that

$$\mathbf{AB} = \mathbf{0}_{100\times 1}.$$

Now it is clear that the equation $\mathbf{AX} = \mathbf{C}$ cannot have a unique solution. Because if $\mathbf{X}$ is a solution then $\mathbf{X} + \mathbf{B}$ is another, different solution:

$$\mathbf{A}(\mathbf{X} + \mathbf{B}) = \mathbf{AX} + \mathbf{AB} = \mathbf{C} + \mathbf{0} = \mathbf{C}.$$

$\square$

Problem 10: (from final exam 2020.)

(a) Give an example of a 2×3 matrix of rank 1.

(b) Let $\mathbf{A}$ be an $m \times n$ matrix of rank 1. Using the definition of rank, show that there exists an $m \times 1$ matrix $\mathbf{X}$ and an $1 \times n$ matrix $\mathbf{Y}$ such that

$$\mathbf{A} = \mathbf{XY}.$$

Solution to (a).

An easy way to do this is just to give a matrix where all the rows are the same non-zero vector, or all the columns are the same non-zero vector. That vector is obviously a basis for the row space (respectively column space), so the dimension of the row space is 1.

For example:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{bmatrix}.$$

Solution to (b).

Because $\mathbf{A}$ has rank 1 the dimension of the column space of $\mathbf{A}$ is 1. This means the column space has a basis consisting of a single $m \times 1$ column vector which we'll denote $\mathbf{X}$. Every column of $\mathbf{A}$ is a scalar multiple of this column vector. Thus there exist scalars y_1 up to y_n such that

$$\mathbf{A} = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ y_1\mathbf{X} & y_2\mathbf{X} & \cdots & y_n\mathbf{X} \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}.$$

This expression can be factored into a matrix product as follows:

$$\mathbf{X} \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} = \mathbf{XY}.$$

□

Problem 11: (from final exam 2020.)

Carefully prove that the two vectors

$$\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

form a basis for the null space of the matrix

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}.$$

You may use standard facts introduced in lectures if you state them when you use them.

Solution.

First we'll check that these vectors actually lie in the null space. Indeed:

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Next we'll note that these vectors are indeed linearly independent. This is because two vectors are linearly dependent if and only if one is a multiple of the other. This is clearly not true here.

Next we'll determine a basis of the null space. The algorithm tell us to first put the matrix in RREF.

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 \end{bmatrix}.$$

$x_1 = s$
$x_2 = -s - t$
$x_3 = t$
$x_4 = s$

So the null space has a basis:

$$\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}.$$

$(s, -s-t, t, s)$

These two vectors can be expressed in terms of the vectors we are given:

$= s(1,-1,0,1) + t(0,-1,1,0)$

$$\begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{2}\left(\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \\ 1 \end{bmatrix}\right)$$

and

$$\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \frac{1}{2}\left(-\begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \\ 1 \\ 1 \end{bmatrix}\right)$$

So we have deduced the two vectors we were given are linearly independent and span the null space (because a known basis can be expressed in terms of them). Thus we deduce they form a basis.

□

Problem 12: (From final exam 2021.)

A polynomial of degree 3 or less is a function $f\colon \mathbb{R} \to \mathbb{R}$ of the form

$$f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

for constants $c_0, c_1, c_2, c_3 \in \mathbb{R}$.

(a) For some given a and b, determine all such polynomials such that $f(1) = a$ and $f(-1) = b$.

(b) Let $S \subset \mathbb{R}^4$ denote the set of 4-tuples (c_0, c_1, c_2, c_3) with the property that the corresponding function $f(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$ satisfies

$$f(-x) = -f(x)$$

for all $x \in \mathbb{R}$. Briefly explain why S is a subspace of $\mathbb{R}^4$ and determine its dimension. (You may assume that if $c_0 + c_1 x + c_2 x^2 + c_3 x^3 = 0$ for all x then $c_0 = c_1 = c_2 = c_3 = 0$.)

Solution to (a).

We need to find all (c_1, c_2, c_3, c_4) such that

$$\begin{aligned} c_0 + c_1 + c_2 + c_3 &= a \\ c_0 - c_1 + c_2 - c_3 &= b \end{aligned}$$

The augmented matrix for this system is:

$$\left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 1 & -1 & 1 & -1 & b \end{array}\right].$$

Now we solve via Gauss-Jordan:

$$\to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 0 & -2 & 0 & -2 & b-a \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 & -\frac{1}{2}b + \frac{1}{2}a \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 0 & 1 & 0 & \frac{1}{2}a + \frac{1}{2}b \\ 0 & 1 & 0 & 1 & \frac{1}{2}a - \frac{1}{2}b \end{array}\right].$$

To write down the general solution we introduce free parameters $c_2 = r$ and $c_3 = s$. The general solution is:

$$\begin{cases} c_0 = \frac{1}{2}a + \frac{1}{2}b - r \\ c_1 = \frac{1}{2}a - \frac{1}{2}b - s \\ c_2 = r \\ c_3 = s. \end{cases}$$

In other words: The set of polynomials of degree 4 satisfying the given conditions is:

$$\left\{ \left(\frac{1}{2}a + \frac{1}{2}b - r\right) + \left(\frac{1}{2}a - \frac{1}{2}b - s\right) x + rx^2 + sx^3; r, s \in \mathbb{R} \right\}.$$

Solution to (b).

The polynomial $c_0 + c_1x + c_2s^2 + c_3x^3$ satisfies the given condition if and only if

$$c_0 + c_1x + c_2x^2 + c_3x^3 = -(c_0 + c_1(-x) + c_2x^2 + c_3(-x)^3)$$

for all x. In other words

$$2c_0 + 2c_2x^2 = 0$$

for all x. This is equivalent to the system of 2 linear equations

$$c_0 = 0 \ , \ \ c_2 = 0.$$

The set S is the set of solutions of this system.

Because this is a homogeneous linear system, the set of solutions is a subspace. Thus S is a subspace. (Some students will check the axioms of a subspace which is also OK.)

We can determine S to be

$$\left\{(0, r, 0, s) \in \mathbb{R}^4; r, s \in \mathbb{R}\right\}.$$

This subspace has a basis $\{(0, 1, 0, 0), (0, 0, 0, 1)\}$ and hence has dimension 2.

$\square$

Problem 13. (From final exam 2022.)

The following matrix $\mathbf{M}$ depends on a parameter $a \in \mathbb{R}$.

$$\mathbf{M} = \begin{bmatrix} 2 & 3 & a \\ 1 & 3 & 1 \\ 1 & 2 & a \end{bmatrix}.$$

(a) For what values of the parameter a is the rank of this matrix less than 3? Justify your answer.

(b) Is the following statement true or false: "When the rank of a 3×3-matrix is less than 3, then the list of rows of the matrix is linearly dependent." Briefly justify using the definitions of these concepts.

(c) Now let a be one of the values you found in Part (a). If possible, express one of the rows of the resulting matrix $\mathbf{M}$ as a linear combination of the other rows.

Solution to (a).

The rank of the matrix is the number of rows in an echelon form of the matrix. So we can answer this by putting the matrix in echelon form.

$$\begin{bmatrix} 2 & 3 & a \\ 1 & 3 & 1 \\ 1 & 2 & a \end{bmatrix} \to \begin{bmatrix} 1 & 2 & a \\ 2 & 3 & a \\ 1 & 3 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & a \\ 0 & -1 & -a \\ 0 & 1 & 1-a \end{bmatrix}$$

Thus we get:

$$\begin{bmatrix} 1 & 2 & a \\ 0 & -1 & -a \\ 0 & 0 & 1-2a \end{bmatrix}.$$

The rank will be less than 3 precisely when $a = 1/2$.

Solution to (b).

The statement is true.

There are different ways of seeing this. One simple way is that the rank is the dimension of the row space of the matrix. If the rows were linearly independent, then they would form a basis for the row space, which would then have dimension 3. This would be a contradiction.

Solution to (c).

So we know set a to $1/2$. The matrix becomes

$$\begin{bmatrix} 2 & 3 & 1/2 \\ 1 & 3 & 1 \\ 1 & 2 & 1/2 \end{bmatrix}$$

We want to solve for a, b and c such that

$$a(2,3,1/2) + b(1,3,1) + c(1,2,1/2) = (0,0,0).$$

This is equivalent to the linear system:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 0 \\ 3 & 3 & 2 & 0 \\ 1/2 & 1 & 1/2 & 0 \end{array}\right].$$

Now we solve:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \\ 3 & 3 & 2 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & -3 & -1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -3 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

We just need 1 solution. Set c to 3. Then $b = -1$. Then $a = -2 * (-1) - 3 = -1$. Thus we deduce:

$$(-1)(2, 3, 1/2) + (-1)(1, 3, 1) + (3)(1, 2, 1/2) = (0, 0, 0).$$

Thus:

$$(2, 3, 1/2) = (-1)(1, 3, 1) + (3)(1, 2, 1/2).$$

$\square$

Problem 14: (From final exam 2022.)

If $\vec{v} = (v_1, \dots, v_n)$ and $\vec{w} = (w_1, \dots, w_n)$ are two n-dimensional vectors, then their scalar product $\vec{v} \cdot \vec{w}$ is defined to be the number

$$\vec{v} \cdot \vec{w} = v_1 w_1 + \dots + v_n w_n.$$

If Z denotes a finite set of vectors $Z \subset \mathbb{R}^n$, denote by $Z^\perp$ the set

$$Z^\perp = \{\vec{v} \in \mathbb{R}^n : \vec{v} \cdot \vec{z} = 0 \text{ for all } \vec{z} \in Z\}.$$

(a) Find a parametrization for $Z^\perp$ in the case $Z = \{(1, -1, 1), (1, 2, 3)\}$.

(b) Show that $Z^\perp$ is a subspace of $\mathbb{R}^n$ for any Z.

(c) Show that

$$\dim(\operatorname{span}(Z)) + \dim(Z^\perp) = n.$$

You may assume standard theory introduced in lectures.

Solution to (a)

We want to find the set of vectors $(v_1, v_2, v_3) \in \mathbb{R}^3$ satisfying the equations

$$\begin{aligned} v_1 - v_2 + v_3 &= 0 \\ v_1 + 2v_2 + 3v_3 &= 0 \end{aligned}$$

We solve these in a standard way.

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 1 & 2 & 3 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 3 & 2 & 0 \end{array}\right]$$

This leads to a parametrization:

$$\left\{(-\frac{5}{3}s, -\frac{2}{3}s, s); s \in \mathbb{R}\right\}.$$

Solution to (b)

If we list the vectors in Z as follows:

$$\begin{aligned} \vec{z}_1 &= (z_{11}, z_{12}, \dots, z_{1n}) \\ &\vdots \\ \vec{z}_k &= (z_{k1}, z_{k2}, \dots, z_{kn}) \end{aligned}$$

then $Z^\perp$ is precisely the set of solutions to the linear system of equations

$$\begin{gathered} z_{11} v_1 + \dots + z_{1n} v_n = 0 \\ \vdots \\ z_{k1} v_1 + \dots + z_{kn} v_n = 0 \end{gathered}$$

This is a homogeneous system of linear equations, so its set of solutions is automatically a subspace, by a standard theorem.

Solution to (c)

If we build a matrix whose rows are the vectors of Z, as follows:

$$\mathbf{M} = \begin{bmatrix} z_{11} & z_{12} & \dots & z_{1n} \\ \vdots & \vdots & \vdots & \vdots \\ z_{k1} & z_{k2} & \dots & z_{kn} \end{bmatrix}$$

then:

- the row space of $\mathbf{M}$ is the span of Z
- the null space of $\mathbf{M}$ is $Z^{\perp}$

and the rank-nullity theorem becomes

$$n = \text{rank}(\mathbf{M}) + \text{nullity}(\mathbf{M}) = \dim(\text{span}(Z)) + \dim(Z^{\perp}).$$

$\square$

Problem 15:

Let V be the subspace of $\mathbb{R}^5$ spanned by the following vectors:

$$\mathbf{v_1} = (1, -2, 0, 0, 3)\ , \quad \mathbf{v_2} = (2, -5, -3, -2, 6)\ , \quad \mathbf{v_3} = (0, 5, 15, 10, 0)\ , \quad \mathbf{v_4} = (2, 1, 15, 8, 6)$$

(i) Find a basis for V. What is $\dim(V)$?

(ii) Extend the basis for V found in part (i) to a basis for $\mathbb{R}^5$.

Solution to (i)

If we apply the standard algorithm to reduce a spanning set to a basis we start by writing down the matrix with these vectors as columns:

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 1 \\ 0 & -3 & 15 & 15 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{bmatrix}$$

Then we do Gaussian elimination:

$$\to \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 5 \\ 0 & -3 & 15 & 15 \\ 0 & -2 & 10 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & -1 & 5 & 5 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now the algorithm tells us to take the vectors from the original list whose corresponding columns in the echelon form matrix have leading entries. So we take:

$$\mathbf{v_1} = (1, -2, 0, 0, 3)\ , \quad \mathbf{v_2} = (2, -5, -3, -2, 6)\ , \quad \mathbf{v_4} = (2, 1, 15, 8, 6)$$

Solution to (ii)

Now we need to extend this basis for V to a complete basis of $\mathbb{R}^5$. The trick is to add a full basis for $\mathbb{R}^5$ onto the end of this list and then repeat the procedure. The result of the algorithm will include the same vectors as before, but will also include a few extra vectors as needed.

So our starting list in this part of the problem is:

$$\mathbf{v_1}\ , \quad \mathbf{v_2}\ , \quad \mathbf{v_4}, \quad \mathbf{e_1}, \quad \mathbf{e_2}, \quad \mathbf{e_3}, \quad \mathbf{e_4}, \quad \mathbf{e_5}$$

where the $\mathbf{e}_i$ refer to the standard basis of $\mathbb{R}^4$. For example $\mathbf{e}_2 = (0, 1, 0, 0, 0)$.

The algorithm proceeds:

$$\begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ -2 & -5 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & -3 & 15 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 8 & 0 & 0 & 0 & 1 & 0 \\ 3 & 6 & 6 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 5 & 2 & 1 & 0 & 0 & 0 \\ 0 & -3 & 15 & 0 & 0 & 1 & 0 & 0 \\ 0 & -2 & 8 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Continuing:

$$\to \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 5 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & -3 & 1 & 0 & 0 \\ 0 & 0 & -2 & -4 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 5 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & -2 & -4 & -2 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -3 & 1 & 0 & -2 \end{bmatrix}$$

Now we can read off the answer. A basis for $\mathbb{R}^5$ which includes our answer for Part (i) is given by the vectors on the list above whose corresponding columns in this matrix have a leading entry. Namely:

$$\mathbf{v_1}\ , \quad \mathbf{v_2}\ , \quad \mathbf{v_4}, \quad \mathbf{e_1}, \quad \mathbf{e_2}.$$

$\square$

This is certainly the correct number of vectors (5). We can check our answer by making sure they are linearly independent using a different method. If our answer is correct the following determinant should be non-zero. (Recall this is a special technique for checking a list of n vectors in $\mathbb{R}^n$ is a basis.)

$$\begin{vmatrix} 1 & 2 & 2 & 1 & 0 \\ -2 & -5 & 1 & 0 & 1 \\ 0 & -3 & 15 & 0 & 0 \\ 0 & -2 & 8 & 0 & 0 \\ 3 & 6 & 6 & 0 & 0 \end{vmatrix} = (-1) \begin{vmatrix} -2 & -5 & 1 & 1 \\ 0 & -3 & 15 & 0 \\ 0 & -2 & 8 & 0 \\ 3 & 6 & 6 & 0 \end{vmatrix} = (-1)^2 \begin{vmatrix} 0 & -3 & 15 \\ 0 & -2 & 8 \\ 3 & 6 & 6 \end{vmatrix} = 3 \begin{vmatrix} -3 & 15 \\ -2 & 8 \end{vmatrix} \neq 0.$$

So our list of vectors is definitely a basis for $\mathbb{R}^5$.

Problem 16. (From final exam 2021.)

Consider the following three vectors in $\mathbb{R}^5$

$$\vec{u} = (1,3,1,3,1),\ \vec{v} = (3,-1,1,-1,1),\ \vec{w} = (-1,2,0,2,0).$$

(a) Compute a basis for $\operatorname{span}\{\vec{u},\vec{v},\vec{w}\}$.

(b) Determine a system of linear equations whose set of solutions is precisely $\operatorname{span}\{\vec{u},\vec{v},\vec{w}\}$.

Solution to (a).

A basis is given by the first two vectors

$$\vec{u} = (1,3,1,3,1),\ \vec{v} = (3,-1,1,-1,1).$$

Solution to (b).

The span is a subspace. Thus we should look for a homogeneous system of equations.

We'll start by determining the set of homogeneous equations containing these two vectors. Consider a general homogeneous equation in $\mathbb{R}^5$:

$$c_1x_1 + c_2x_2 + c_3x_3 + c_4x_4 + c_5x_5 = 0.$$

The vector $(1,3,1,3,1)$ will solve this equation iff:

$$c_1 + 3c_2 + c_3 + 3c_4 + c_5 = 0.$$

And the vector $(3,-1,1,-1,1)$ will solve this equation iff:

$$3c_1 - c_2 + c_3 - c_4 + c_5 = 0.$$

The augmented matrix corresponding to these two equations is:

$$\left[\begin{array}{ccccc|c} 1 & 3 & 1 & 3 & 1 & 0 \\ 3 & -1 & 1 & -1 & 1 & 0 \end{array}\right].$$

Solving this:

$$\to \left[\begin{array}{ccccc|c} 1 & 3 & 1 & 3 & 1 & 0 \\ 0 & -10 & -2 & -10 & -2 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & 3 & 1 & 3 & 1 & 0 \\ 0 & 1 & 1/5 & 1 & 1/5 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{ccccc|c} 1 & 0 & 2/5 & 0 & 2/5 & 0 \\ 0 & 1 & 1/5 & 1 & 1/5 & 0 \end{array}\right]$$

This gives a general solution:

$$\begin{cases} c_1 = -2/5r - 2/5t \\ c_2 = -1/5r - s - 1/5t \end{cases}$$

We get three equations by setting 1 variable to 1 and the others to zero:

$$\begin{cases} -2/5x_1 - 1/5x_2 + x_3 = 0 \\ -x_2 + x_4 = 0 \\ -2/5x_1 - 1/5x_2 + x_5 = 0 \end{cases}$$

$\square$

Problem 17:

Consider a list of non-zero vectors in some $\mathbb{R}^m$.

$$\mathbf{v}_1\,,\ \mathbf{v}_2\,,\ \ldots\ ,\ \mathbf{v}_k.$$

Assume this list has the property that the position of first non-zero entry of $\mathbf{v}_j$ is further to the right than the position of the first non-zero entry of $\mathbf{v}_{j-1}$ for all $1 < j \leq k$.

Prove that this list is linearly independent.

Solution

The clearest way to organize this proof is to use induction on the number k. The base case $k = 1$ is clear because a single vector is a linearly independent list.

(*The induction step.*)

Our induction assumption is that when a list has this property and has length less than k then it is linearly independent.

Next we consider a list with this property of length k.

$$\mathbf{v}_1\,,\ \mathbf{v}_2\,,\ \ldots\ ,\ \mathbf{v}_k.$$

Our problem is to explain that the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k = \mathbf{0} \quad (\star)$$

only has the trivial solution.

Let p denote the position of the first non-zero entry of $\mathbf{v}_1$. Because the first non-zero position of the vectors in this list moves strictly to the right, we know that all the vectors on this list except for the vector $\mathbf{v}_1$ has a zero in position p. Thus the position p entry of the LHS of equation $(\star)$ is precisely c_1, while of course the position p entry of the RHS is exactly zero. We deduce

$$c_1 = 0.$$

Thus the equation $(\star)$ above implies

$$c_2\mathbf{v}_2 + \ldots + c_k\mathbf{v}_k = \mathbf{0} \quad (\star\star)$$

But the list

$$\mathbf{v}_2\,,\ \ldots\ ,\ \mathbf{v}_k.$$

has length shorter than k, and still has the property that the first non-zero position moves strictly to the right. Thus by the induction assumption this list is linearly independent. Thus the equation $(\star\star)$ implies in addition

$$c_2 = \ldots = c_k = 0.$$

Thus the given list is linearly independent and the induction is complete.

□

Problem 18:

Let V be a subspace of $\mathbb{R}^n$ and assume we have two different bases for it:

$$B = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\} \subset V \ , \ B' = \{\mathbf{w}_1, \ldots, \mathbf{w}_n\} \subset V.$$

Prove that $m = n$.

Hints:

You probably want to show that you can get from B to B' via a sequence of intermediate bases B_i:

$$B = B_0 \to B_1 \to B_2 \to \ldots \to B_m = B'.$$

- All these bases will contain the same number m of vectors.
- The basis B_i will contain some of the vectors from B and the first i vectors of B'.
- B_{i+1} will be obtained from B_i by replacing one of the vectors coming from B with the vector w_{i+1}. Which vector will you replace? Why is the result a basis?

Solution

We'll use induction to prove: There exists a basis B_i of V of length m which starts with $m - i$ vectors from B and finishes with the first i vectors of B'. The induction will be on the number i.

(*Base case.*) The base case is the case $i = 0$. In this case B_0 is just B.

(*The induction step.*)

We assume we have such a basis B_{i-1}. We'll explain how to construct the basis B_i from it.

For the discussion assume that the vectors on the list B_{i-1} are given as follows. The numbers $j_1, \ldots, j_{m-i+1}$ record which vectors from B are on the list B_{i-1}:

$$\mathbf{v}_{j_1} \ , \ \ldots \ , \ \mathbf{v}_{j_{m-i+1}} \ , \ \mathbf{w}_1 \ , \ \ldots , \ \mathbf{w}_{i-1}.$$

Now add $\mathbf{w}_i$ to the end of this list B_{i-1}. Because $\mathbf{w}_i \in V$ and B_{i-1} is a basis for V by the induction assumption, we know that there is some way to express $\mathbf{w}_i$ using the vectors in the list B_{i-1} and we know that not all of these coefficients are zero (because $\mathbf{w}_i \neq 0$):

$$\mathbf{w}_i = c_1 \mathbf{v}_{j_1} + \ldots + c_{m-i+1} \mathbf{v}_{j_{m-i+1}} + d_1 \mathbf{w}_1 + \ldots + d_{i-1} \mathbf{w}_{i-1}. \quad (\star)$$

Next note that at least one $c_r \neq 0$. (Because we know that the set B' is a basis, so it is linearly independent. If all the $c_j = 0$ then this equation would imply B' is linearly dependent.)

We will remove the vector $\mathbf{v}_{j_r}$ from this list and the result will be our next basis B_i.

We must check that this list B_i we have just constructed is indeed a basis for V.

First note that we can rearrange the equation $(\star)$ above to write the following equation. (The symbol $\widehat{\ }$ we use below is mathematical shorthand for "omit this term when writing down following this pattern".)

$$\mathbf{v}_{j_r} = -\frac{c_1}{c_r}\mathbf{v}_{j_1} + \ldots + \widehat{-\frac{c_1}{c_r}\mathbf{v}_{j_r}} + \ldots + -\frac{c_{m-i+1}}{c_r}\mathbf{v}_{j_{m-i+1}} - \frac{d_1}{c_r}\mathbf{w}_1 + \ldots - \frac{d_{i-1}}{c_r}\mathbf{w}_{i-1} + \frac{1}{c_r}\mathbf{w}_i.$$

This equation shows that this new list B_i does indeed span V because it is possible to express every one of the vectors from the basis B_{i-1} in terms of it.

The last thing we have to do is explain why B_i is linearly independent. To test this we assume that we have numbers such that

$$a_1\mathbf{v}_{j_1} + \ldots + \widehat{a_r\mathbf{v}_{j_r}} + \ldots + a_{m-i+1}\mathbf{v}_{j_{m-i+1}} + b_1\mathbf{w}_1 + \ldots + b_i\mathbf{w}_i = \mathbf{0}.$$

We need to show all these numbers are zero.

If we substitute $(\star)$ into this equation we eliminate the new vector $\mathbf{w}_i$. Collecting terms we deduce:

$$\begin{aligned}
\mathbf{0} = \quad & (a_1 + b_ic_1)\mathbf{v}_{j_1} \\
+ \quad & (a_2 + b_ic_2)\mathbf{v}_{j_2} \\
\vdots \quad & \\
+ \quad & b_ic_r\mathbf{v}_{j_r} \\
\vdots \quad & \\
+ \quad & (a_{m-i+1} + b_ic_{m-i+1})\mathbf{v}_{j_{m-i+1}} \\
+ \quad & (b_1 + b_i)\mathbf{w}_1 \\
+ \quad & (b_2 + b_i)\mathbf{w}_2 \\
\vdots \quad & \\
+ \quad & (b_{i-1} + b_i)\mathbf{w}_{i-1}.
\end{aligned}$$

Because the vectors of B_{i-1} are linearly independent and $c_r \neq 0$, we deduce $b_i = 0$. Thus this equation becomes:

$$\begin{aligned}
\mathbf{0} = \quad & a_1\mathbf{v}_{j_1} \\
+ \quad & a_2\mathbf{v}_{j_2} \\
\vdots \quad & \\
+ \quad & a_{m-i+1}\mathbf{v}_{j_{m-i+1}} \\
+ \quad & b_1\mathbf{w}_1 \\
+ \quad & b_2\mathbf{w}_2 \\
\vdots \quad & \\
+ \quad & b_{i-1}\mathbf{w}_{i-1}.
\end{aligned}$$

Thus we deduce that

$$a_1 = a_2 = \ldots = a_{m-i+1} = b_1 = \ldots = b_{i-1} = 0.$$

$\square$